

VoxelScan Inspection Report

Scan vol_001 | generated 2026-07-25 12:59

SIMULATED volumetric scan data - analysis workflow demo, not a real asset and not calibrated acoustics.

Scan metadata

Scan ID	vol_001
Volume shape (z,y,x)	128 x 128 x 128 voxels
Data type	Simulated volumetric scan (acoustic-style amplitude)
Simulated noise (std)	4.9 (difficulty 0.05)
Detections (pre-review)	5
Confirmed / rejected	5 / 0 (reviewer: auto)

Findings summary

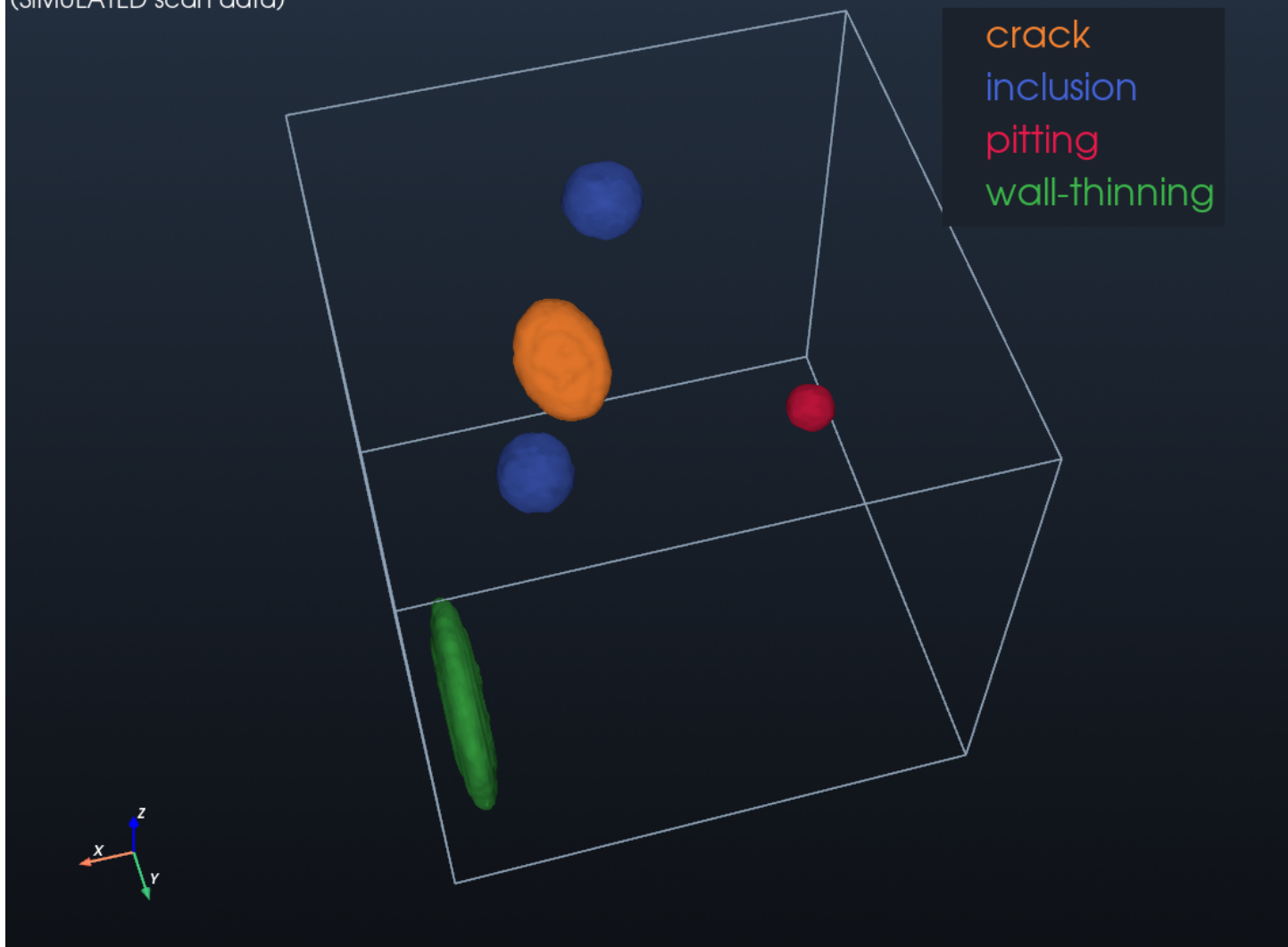
Automated analysis of simulated scan vol_001, followed by human-in-the-loop review, confirmed 5 defects: 1 crack, 2 inclusions, 1 pitting, 1 wall-thinning. By severity index (equivalent size x intensity deviation), 4 rank high, 1 medium, and 0 low. The most significant confirmed feature is a inclusion centred at (z=64, y=29, x=114) with an equivalent diameter of 17.4 voxels and a peak intensity deviation of +64 from the local baseline. All figures are measurements from automated detection on simulated volumetric scan data; this report characterises the analysis workflow and does not model acoustic physics or represent a real asset.

Confirmed defects

ID	Type	Loc (z,y,x)	Size (vox)	Eq.dia	Peak d	Severity
det2	inclusion	64, 29, 114	2735	17.4	+64	high (1103)
det5	wall-thinning	117, 101, 37	11677	28.1	-32	high (909)
det4	inclusion	92, 79, 89	2744	17.4	+51	high (887)
det3	crack	76, 29, 60	4165	20.0	+43	high (859)
det1	pitting	14, 39, 32	978	12.3	-43	medium (530)

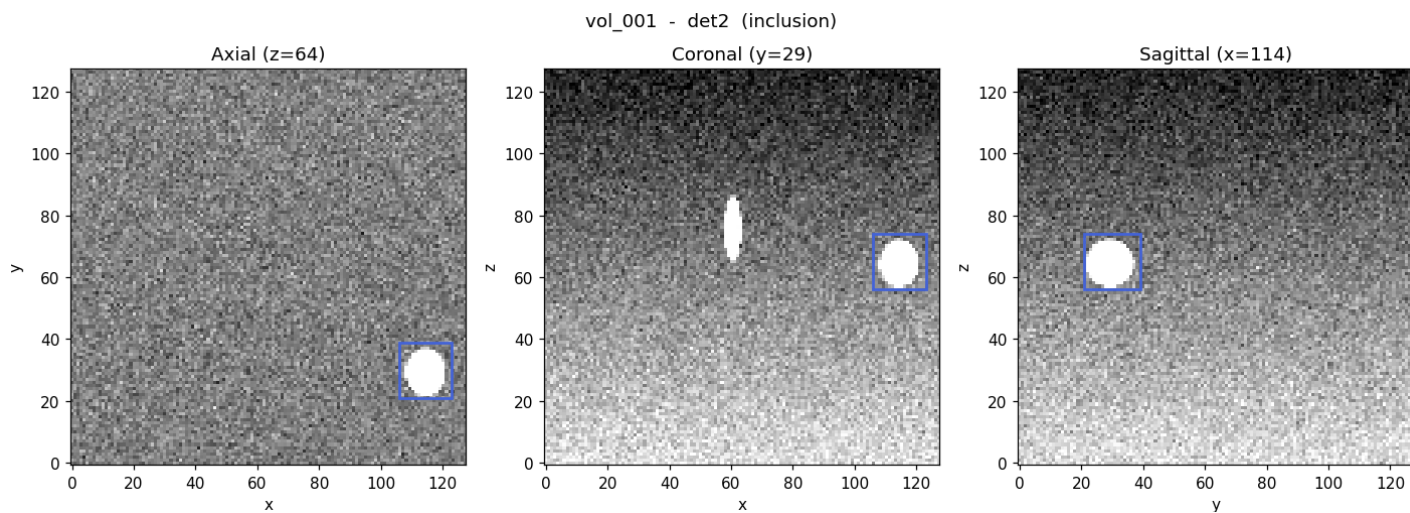
3D defect map

vol_001 - 5 detected defects
(SIMULATED scan data)



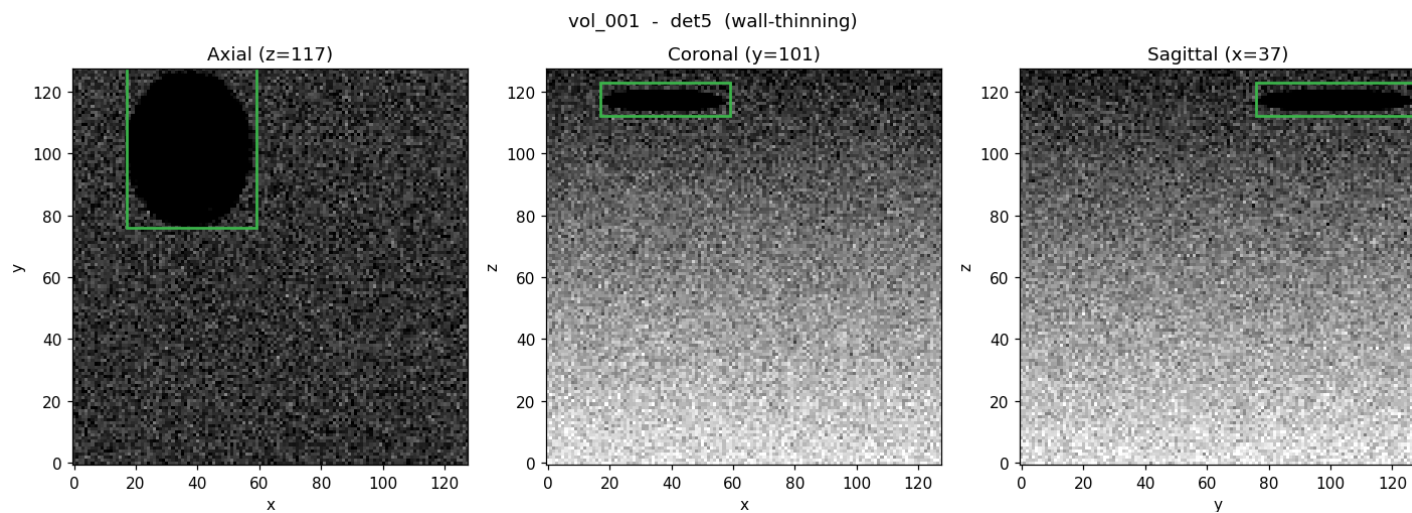
Detected defects rendered as colour-coded isosurfaces inside the scanned block (translucent). Colours indicate defect type. Simulated data.

Slice views - det2 (inclusion)



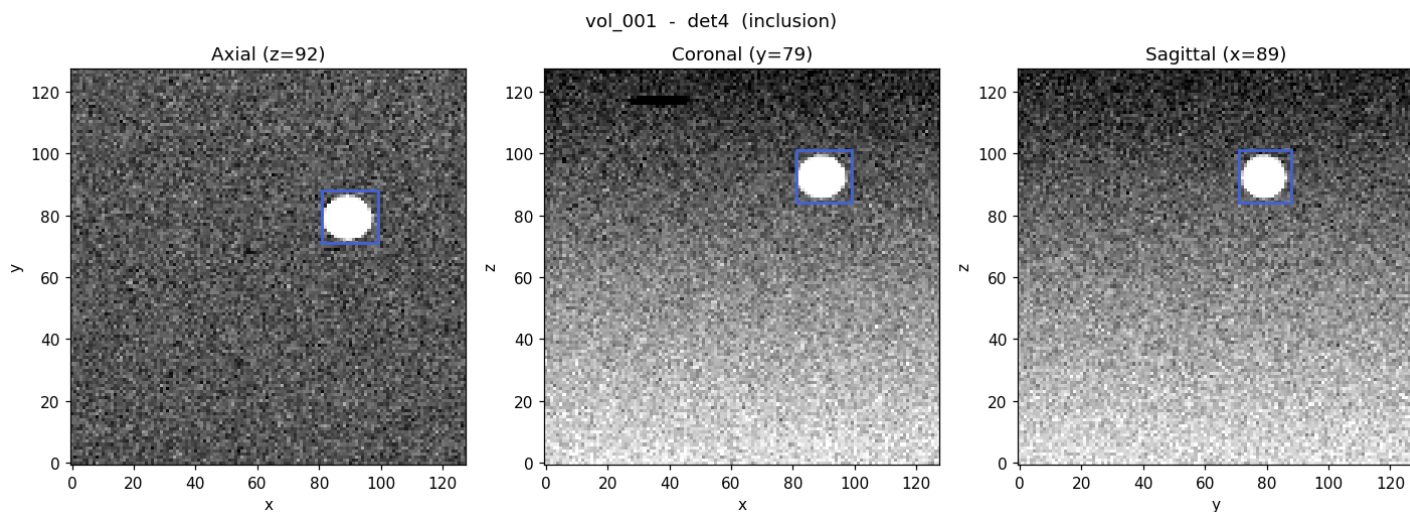
Axial / coronal / sagittal slices through the inclusion at $(z,y,x)=(64, 29, 114)$. Green box = detected bounding box. Equivalent diameter 17.4 voxels, severity high.

Slice views - det5 (wall-thinning)



Axial / coronal / sagittal slices through the wall-thinning at (z,y,x)=(117, 101, 37). Green box = detected bounding box. Equivalent diameter 28.1 voxels, severity high.

Slice views - det4 (inclusion)



Axial / coronal / sagittal slices through the inclusion at (z,y,x)=(92, 79, 89). Green box = detected bounding box. Equivalent diameter 17.4 voxels, severity high.